

GOVERNMENT ENGINEERING COLLEGE, RAJKOT

GENERAL DEPARTMENT

Subject: **Discrete Mathematics & Graph Theory (BE04000261)**

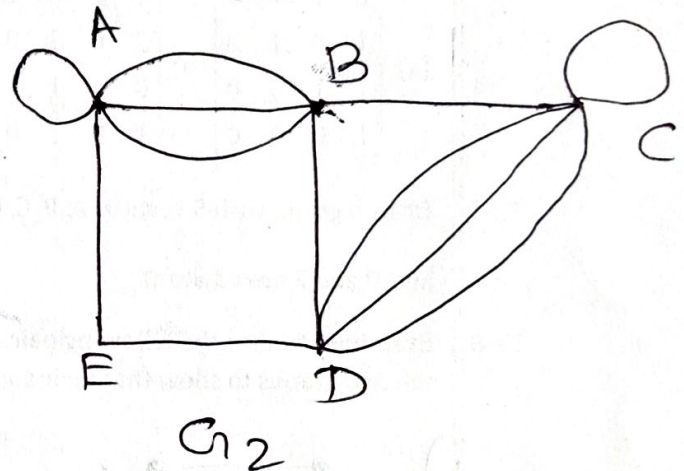
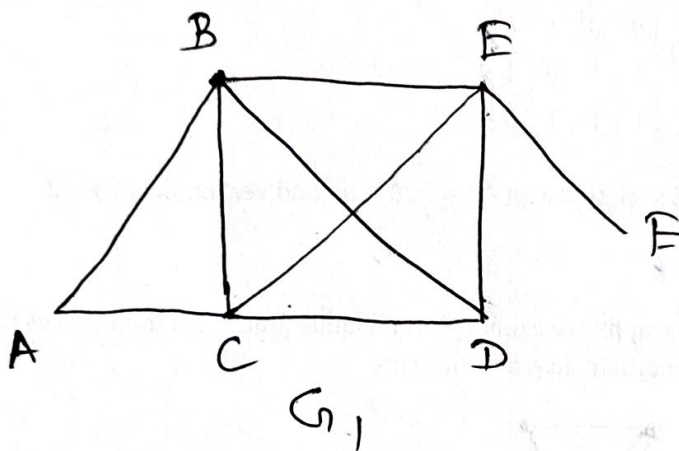
Semester: 4th Branch: **CE** Tutorial-05 (CO-5) Term Date: 01/2/26 to 06/5/2026

Ex-1 Draw the diagram of each of the following graphs $G = (V, E)$

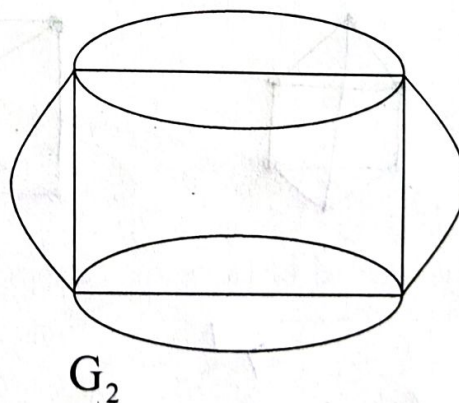
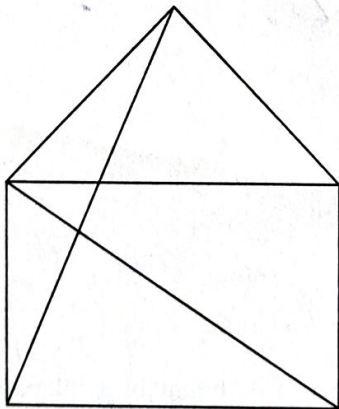
(a) $V = \{1, 2, 3, 4, 5\}$, $E = \{(1, 2), (1, 4), (1, 5), (2, 3), (3, 4), (4, 4)\}$

(b) $V = \{1, 2, 3, 4, 5, 6\}$, $E = \{(1, 2), (1, 4), (1, 4), (2, 3), (2, 5), (3, 5)\}$

Ex-2 Find the number of vertices, the number of edges and the degree of each vertex in the following graph. Verify also the handshaking theorem in each case.

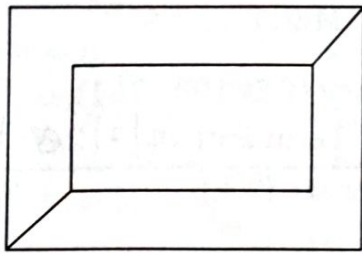


Ex-3 Find the in-degree and out-degree of each vertex of each of the following directed graphs. Also verify that the sum of the in-degree (or out degree) equals to the number of edges.

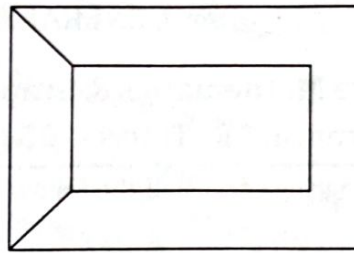


Ex-4 Draw all the sub graphs of K_3 containing at least one vertex.

Ex-5 Determine whether the graphs G_1 and G_2 are isomorphic? Justify.



G_1



G_2

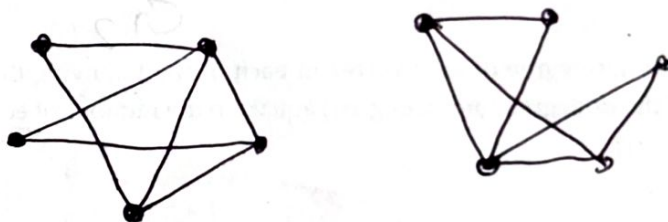
Ex-6 Draw the graphs represented by the following adjacency matrices

(a) $\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

Ex-7 Draw a graph with 5 vertices A, B, C, D, E such that $\deg(A) = 3$, B is an odd vertex, $\deg(C) = 2$ and D and E are adjacent.

Ex-8 Examine whether the following pairs of graphs are isomorphic. If isomorphic, label the vertices of the two graphs to show that their adjacency matrices are the same.

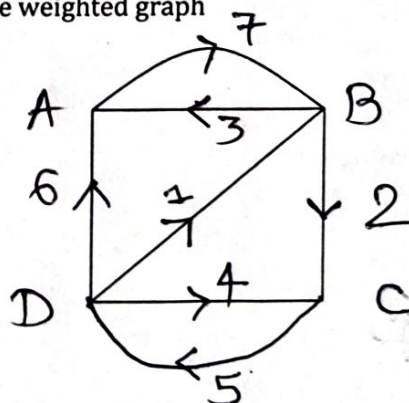
i)



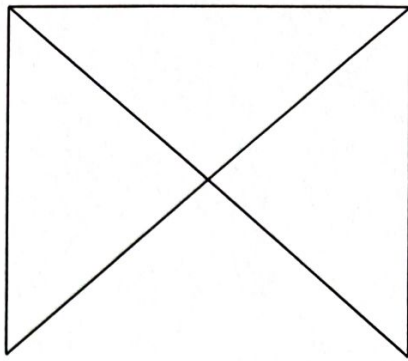
ii)



Ex-9 Find the shortest distance matrix and corresponding shortest path matrix for all pairs of vertices in the weighted graph



Ex-10 Draw all the spanning trees of the graphs G shown in below.



G

Ex-11 Suppose the preorder and inorder traversals of a binary tree T yield the following sequence.

Preorder: GAQACKFPDERH

Inorder: QBKCFAGPEDHR

- (a) Draw the unique Binary tree
- (b) Find (i) depth d of tree(ii) the descendants of B
- (c) List the terminal nodes of T.

Ex-12 Define Directed Tree, leaf and branch nodes with an example.

Ex-13 Traverse the following tree in inorder, preorder and postorder

